PCM-D50

Linear PCM Recorder

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SONY

Outline

The latest addition to Sony's family of portable audio recorders, the **PCM-D50** recorder adopts many of its features and style from the highly acclaimed PCM-D1 model, including rugged construction and cool design. It's the ideal choice for making live musical, wild tracks and field recordings.

The PCM-D50 is a 96kHz/24-bit recorder fitted with two-position (X-Y or Wide) stereo microphones, 4GB of internal flash memory and a Memory Stick Pro-HG Duo™ slot for additional storage. Operating on four AA alkaline batteries, it provides 14 hours of record operation at 44.1kHz 16-bit recording. Unique features such as a two-position stereo microphone, a pre-record buffer that records five seconds of audio before hitting the record button, digital pitch control and A/B segment/repeat offer great portable possibilities for songwriters, musicians and journalists.

To complement the PCM-D50 recorder, new optional accessories include a remote control, tripod stand and microphone windscreen.

Features

Superb Recording Quality
The PCM-D50 is a 96kHz/24-bit linear recorder that records in standard .WAV file format and surpasses standard DAT sound quality. A low signal-to-noise ratio provides virtually noise-free recording quality.

Expandable Memory
With a 4GB built-in memory, the PCM-D50 records up to six hours when recording in 44.1kHz 16-bit CD quality mode. The Memory Stick Pro-HG Duo Slot provides up to 4GB memory expandability.

Built-in Microphones
The PCM-D50 recorder comes equipped with a high sensitivity, built-in, two-position electret condenser stereo microphone with flexible rotation for either X-Y or Wide stereo positions.

Versatile Recording Functions
The Dual Digital Limiter feature helps prevent distortion, a pre-record buffer records five seconds of audio before hitting the record button and Super Bit Mapping technology helps achieve wide, dynamic range.

Playback Functions
The PCM-D50 recorder provides MP3 playback, a special Digital Pitch Control feature that slows down playback without changing pitch and an A/B segment/repeat feature that allows you to mark and repeat segments.

PC/Macintosh Compatible
The PCM-D50 utilises a simple drag and drop file transfer via a high-speed USB connection.

Rugged Construction
Constructed of lightweight metal (aluminium), the PCM-D50 recorder is built to withstand the demands of professional applications.

PCM-D50 Specifications	
Built in Microphones	Electret condenser microphones offering X-Y or Wide stereo positions. High sensitivity (-35.0dB /Pa 1 kHz) (Typical) Maximum input level 120dB SPL Noise level 20.0dB SPL(A) (Typical)
Recording Media	Built-in 4GB flash memory,Memory Stick Pro-HG Duo™ (optional), Memory Stick Pro Duo (High Speed) (optional), Stereo Recording
Sampling Rates	22.05kHz, 44.1kHz, 48kHz and 96kHz
Quantisation	16-bit linear, 24-bit linear
Recording Format	.WAV
Playback Format	.WAV / MP3
Maximum Record Time using Internal 4GB Flash Memory	22kHz 16-bit Recording Mode: 12 hours and 55 minutes; 44.1kHz 16-bit Recording Mode: 6 hours and 25 minutes; 44.1kHz 24-bit Recording Mode: 4 hours and 15 minutes; 48kHz 16-bit Recording Mode: 5 hours and 55 minutes; 48kHz 24-bit Recording Mode: 3 hours and 55 minutes; 96kHz 16-bit Recording Mode: 2 hours and 55 minutes; 96kHz 24-bit Recording Mode: 1 hour and 55 minutes
Frequency Response (Line Input to Line Output)	For Fs = 22.05kHz: Frequency Response = 20Hz to 10kHz (0 to -2dB) For Fs = 44.1kHz: Frequency Response = 20Hz to 20kHz (0 to -2dB) For Fs = 48kHz: Frequency Response = 20Hz to 22kHz (0 to -2dB) For Fs = 96kHz: Frequency Response = 20Hz to 40kHz (0 to -2dB)
Signal-to-Noise Ratio (Line Input to Line Output)	93dB or greater (1kHz IHF-A) when set to 24-bit
Total Harmonic Distortion (Line Input to Line Output)	0.01% or below (1kHz, 22kHz LPF)
Wow and Flutter	Below measurable limit (less than +/-0.001%W.Peak)
Microphone Input (Stereo Mini Jack)	Input impedance: 22k ohm, Rated input level: 2.5 mV; Minimum input level: 0.7 mV. Supports external mic plug-in power.
Headphone Output (Stereo Mini Jack)	Rated output level: 400 mV; Maximum output level: 25 mW + 25 mW or more; Load impedance: 16 ohms
Line Input (OPT)	Analogue input: Input impedance: 40K ohms; Rated input level: 2.0V; Minimum input level: 450 mV. Optical digital input: Input level -24.5dBm to -14.5dBm. Absorption wavelength: 630nm to 690nm
Line Output (OPT)	Analogue output: Output impedance: 220 ohms, Rated output level: 1.7V load impedance: 22k ohms. Optical digital output: Output level -21dBm to -15dBm. Emission wavelength: 630nm to 680nm.
USB Connection	High-speed USB, mass storage class
Memory Stick® Slot	Memory Stick Pro-HG Duo™ media, NOTE: Standard Memory Stick® media, not supported
Power Requirements	DC in 6V. Four AA size Alkaline batteries (supplied); Four AA NiMH Rechargeable batteries (optional)
Power Consumption	0.75W
Approx. Battery Life	14 Hours @44.1/16 recording or 12 Hours @ 96/24 recording
Dimensions (W x H x D)	Approximately 73 mm x 156 mm x 33 mm not including projecting parts and controls
Weight	Approximately 365.14 g (including batteries)
Supplied Accessories	Sound Forge® Audio Studio LE Software CDROM, USB Cable, 4x Alkaline Batteries (AA size), Operating Instructions

XLR-1 / Mic Adapter Option for PCM-D1 and PCM-D50 Recorders

Description

The XLR-1 is ideal for portable recording applications with professional microphones requiring 48 volt phantom power and XLR connections. The XLR-1 adapter has two balanced XLR inputs and switchable 48 volt phantom power. It can be mounted onto the bottom of the PCM-D1 or PCM-D50 or on-the-side (using supplied mounting hardware).

Highlights

The optional XLR-1 adapter has two balanced XLR inputs and switchable 48 volt phantom power.

Features

- Incorporates high quality microphone balancing transformers
- Internal battery holder with four AA batteries
- Flexible mounting onto PCM-D1 and PCM-D50 recorders

Benefits

- The XLR-1 audio signal path is passive to ensure high sound quality
- Provides 48 volt phantom powering for use with professional condenser microphones
- Supplied mounting accessories allow secure mounting

XLR-1 Specifications	
Frequency Range	20 to 50,000 Hz (+0, -2dB)
Input Connectors	XLR-31-11C type connector (3-pin) x 2
Input Impedance	600 ohms
Quantisation	16-bit linear, 24-bit linear
Rated Input Level	-50dBm
Maximum Input Level	+4dBm
Output Connector	Stereo mini jack
Output Load Impedance	600 ohms
Phantom Power Supply	DC 48V ± 0.5V; 10 mA maximum
Battery Requirements	Four AA alkaline batteries (supplied) or Four AA nickel metal hydride rechargeable batteries (not supplied)
Battery Life	Approximately 40 hours (when using 2 mA + 2mA microphones and alkaline AA batteries)
Dimensions (W x H x D)	Approximately 71 mm x 38 mm x 132 mm not including projecting parts and controls
Weight	Approximately 524.48 g (for XLR-1 main unit, including 4 AA batteries); Approximately 184.28 g (for base plate and spacer plate)

PCM-D50 Optional Accessories

RMPCM1 / Remote Commander® Unit Remote Control

VCTPCM1 / Tripod Stand

ADPCM1 / Wind Screen

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